CHAPTER

18

CMOS Processing Technology

With the high-order effects of MOS devices covered in Chapter 17, we now study the fabrication of CMOS technologies. A solid understanding of device processing proves essential in the design and layout of ICs because many limitations imposed on the performance of circuits are related to fabrication issues. Furthermore, today's semiconductor technology demands that process engineers and circuit designers interact regularly so as to understand each other's needs, necessitating a good knowledge of each discipline.

In this chapter, we deal with the processing technology of CMOS devices, aiming to provide a simple view of the fabrication steps and their relevance to circuit design and layout. We begin with a brief description of basic fabrication steps, such as wafer processing, photolithography, oxidation, ion implantation, deposition, and etching. Next, we study the fabrication sequence of MOS transistors in detail. Finally, we describe the processing of passive devices and interconnections.

18.1 ■ General Considerations

Before delving into a detailed study of fabrication, it is instructive to consider the basic structure of NMOS and PMOS transistors and predict the required processing steps. As shown in Fig. 18.1, a *p*-type substrate (wafer) serves as the foundation upon which *n*-wells, source/drain regions, gate dielectric, polysilicon, *n*-well and substrate ties, and metal interconnects are built. Considering both the side view and the top view, we may raise the following questions: (1) How are various regions defined so accurately? For example, how is a gate polysilicon line with a minimum dimension of 0.25 μm fabricated while maintaining a distance of 0.25 μm from another polysilicon line? (2) How are the *n*-wells and S/D regions built? (3) How are the gate oxide and polysilicon fabricated? (4) How are the gate oxide and polysilicon *aligned* with the S/D regions? (5) How are the contact windows created? (6) How are the metal interconnect layers deposited?

Modern CMOS technologies involve more than 200 processing steps, but for our purposes, we can view the sequence as a combination of the following operations: (1) wafer processing to produce the proper type of substrate; (2) photolithography to precisely define each region; (3) oxidation, deposition, and ion implantation to *add* materials to the wafer; and (4) etching to *remove* materials from the wafer. Many of these steps require "heat treatment," i.e., the wafer must undergo a thermal cycle inside a furnace.

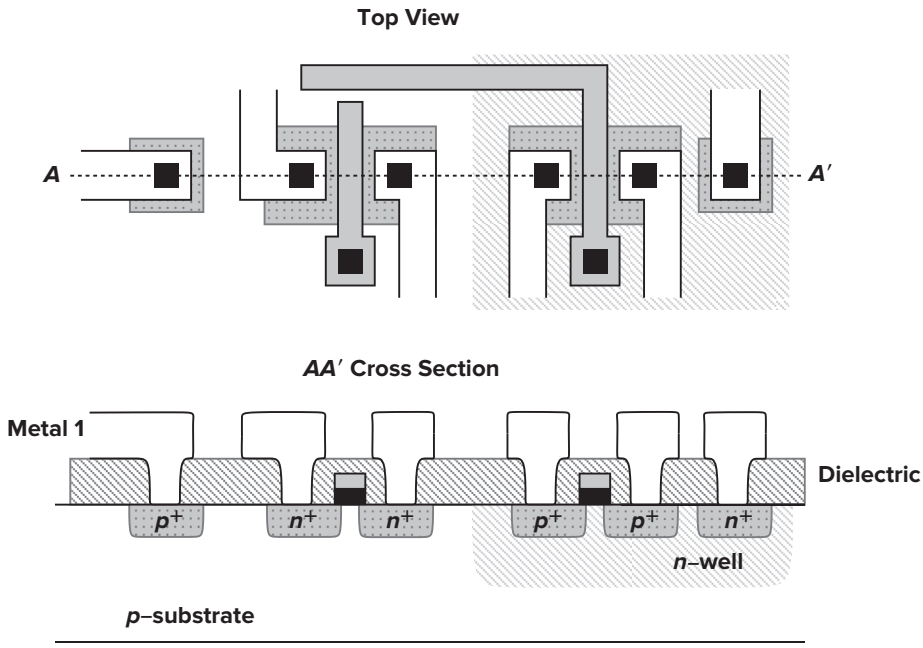


Figure 18.1 Side view and top view of MOS devices.

In semiconductor processing and characterization, we often refer to the "sheet resistance" of a layer. The total resistance of a rectangular bar is $R = \rho L/(W \cdot t)$, where ρ is the resistivity of the material and L, W, and t denote the length, width, and thickness of the bar, respectively. In integrated circuits, the resistivity and thickness of the layers are set by fabrication materials and processing steps and cannot be changed in the layout. The quantity $R_\square = \rho/t$ is thus defined as the sheet resistance, combining two constants of the technology. Since $R = R_\square$ for $W = L$, i.e., for a square geometry, we express $R_\square$ in terms of ohms per square. For example, for a sheet resistance of 10 $\Omega/\square$, a geometry with $W = 2$ μm and $L = 20$ μm has a resistance of $R = 10\ \Omega/\square \times (20/2) = 100\ \Omega$. In fact, we may say "this line is 10 squares long," meaning that $L/W = 10$ and $R = 10R_\square$.

18.2 ■ Wafer Processing

The starting wafer in a CMOS technology must be created with a very high quality. That is, the wafer must be grown as a single-crystal silicon body having a very small number of "defects," e.g., dislocations in the crystal or unwanted impurities. Furthermore, the wafer must contain the proper type and level of doping so as to achieve the required resistivity.

This is accomplished by the "Czochralski method," whereby a seed of crystalline silicon is immersed in molten silicon and gradually pulled out while rotating. As a result, a large single-crystal cylindrical "ingot" is formed that can be sliced thin into wafers. The diameter of the wafer has scaled up with new technology generations, exceeding 30 cm (12 in) today. Note that dopants are added to the molten silicon to obtain the desired resistivity. The wafers are then polished and chemically etched, thereby removing damages on the surface that are created during slicing. In most CMOS technologies, the wafer has a resistivity of 0.05 to 0.1 Ω·cm and a thickness of approximately 500 to 1,000 μm (which is reduced to a few hundred microns after all of the processing steps).

18.3 ■ Photolithography

Photolithography, or simply lithography, is the first step in transferring the circuit layout information to the wafer. As shown in the top view of Fig. 18.1 and explained in Chapter 19 in more detail, the layout consists of polygons representing different types of "layers," e.g., n-well, S/D regions, polysilicon, contact windows, etc. For fabrication purposes, we decompose the layout into these layers. For example, the layout of Fig. 18.1 can be viewed as the five different layers shown in Fig. 18.2, each of which must be created on the wafer with a very high precision. Note that the "active" (or "diffusion") layer includes the source/drain regions and the p^+ and n^+ openings serving as the substrate and well ties.

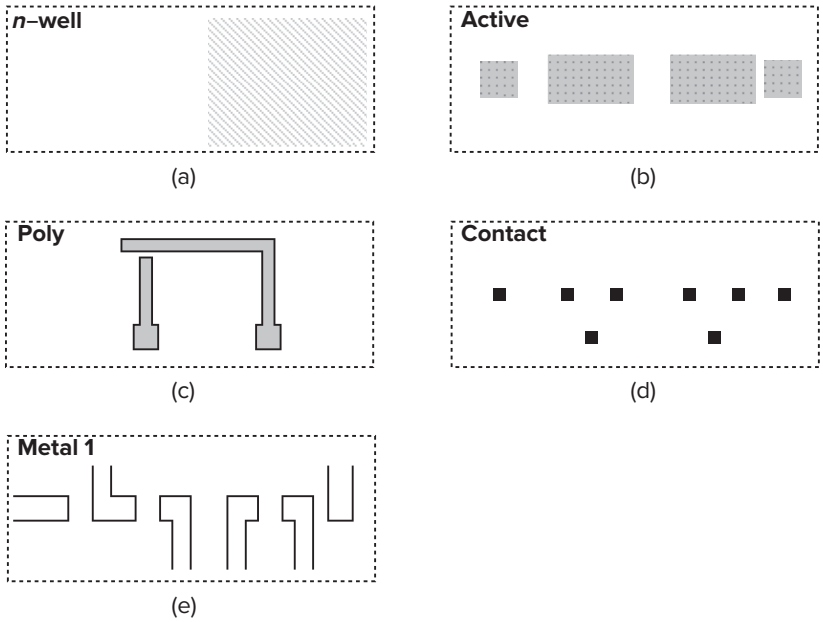


Figure 18.2 Layers comprising the structures of Fig. 18.1.

To understand how a layer is transferred from the layout to the wafer, let us consider the n-well pattern of Fig. 18.2(a) as an example. This pattern is "written" to a transparent glass "mask" by a precisely controlled electron beam [Fig. 18.3(a)]. Also, as depicted in Fig. 18.3(b), the wafer is covered by a thin layer of "photoresist," a material whose etching properties change upon exposure to light.[1] Subsequently, the mask is placed on top of the wafer and the pattern is projected onto the wafer by ultraviolet (UV) light [Fig. 18.3(c)]. The photoresist "hardens" in the regions exposed to light and remains "soft" under the opaque rectangle. The wafer is then placed in an etchant that dissolves the "soft" photoresist area, thereby exposing the silicon surface [Fig. 18.3(d)]. Now, an n-well can be created in the exposed area. We call this set of operations a lithography sequence.

In summary, the sequence associated with the lithography of each layer involves one mask and three processing steps: (1) cover wafer with photoresist; (2) align mask on top and expose to light; (3) etch exposed photoresist. The example of Fig. 18.2 therefore requires at least five masks and hence five lithography sequences.

[1] In practice, a thin layer of oxide is grown before depositing the photoresist to protect the surface.

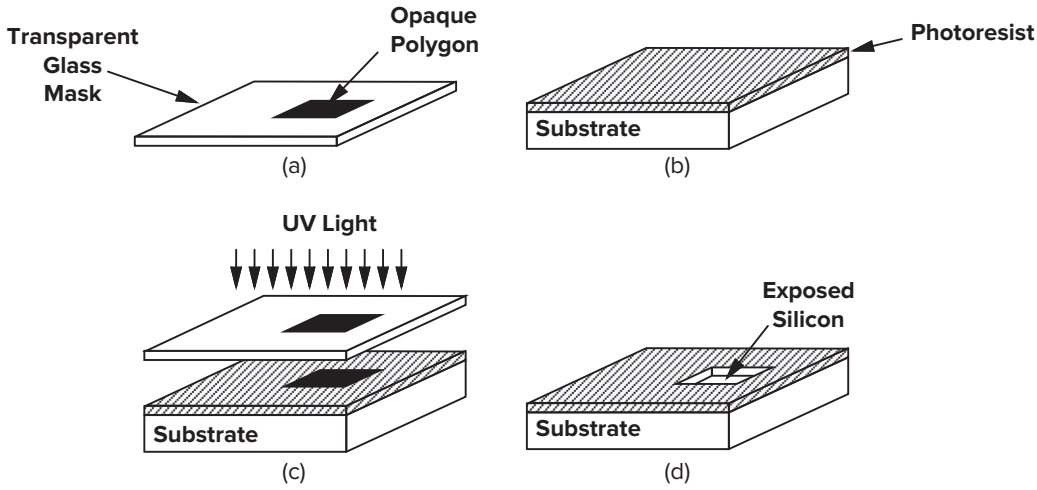


Figure 18.3 (a) Glass mask used in lithography; (b) coverage of wafer by photoresist; (c) selective exposure of photoresist to UV light; (d) exposed silicon after etching.

We should mention that two types of photoresists are used in processing. A "negative" photoresist hardens in the areas exposed to light, and a "positive" photoresist hardens in the areas not exposed to light. As explained later in this chapter, both types prove useful in fabrication.

The number of masks in a process heavily affects the overall cost of fabrication, eventually influencing the unit price of the chip. This is so for two reasons: each mask costs tens of thousands of dollars, and, owing to the necessary precision, lithography is a slow and expensive task. In fact, CMOS technology originally became attractive by virtue of the relatively small number of masks—about seven—that it required. Although in modern CMOS processes this number is around 30, the cost of each IC has nonetheless remained low because both the number of transistors per unit area and the size of the wafer have steadily increased.

18.4 ■ Oxidation

A unique property of silicon is that it can produce a very uniform oxide layer on the surface with little strain in the lattice, allowing the fabrication of gate-oxide layers as thin as a few tens of angstroms (only several *atomic* layers). In addition to serving as the gate dielectric, silicon dioxide can act as a protective coating in many steps of fabrication. Also, in areas between the devices, a thick layer of SiO_2, called the "field oxide" (FOX) is grown, providing the foundation for interconnect lines that are formed in subsequent steps (Fig. 18.4).

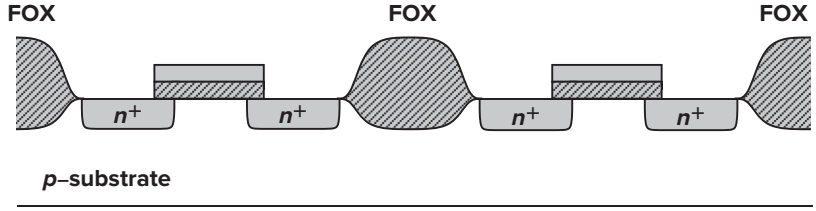


Figure 18.4 Field oxide.

Silicon dioxide is "grown" by placing the exposed silicon in an oxidizing atmosphere such as oxygen at a temperature around $1,000^{\circ}$C. The rate of growth depends on the type and pressure of the atmosphere, the temperature, and the doping level of the silicon.

The growth of the gate oxide is a very critical step in the process. Since the oxide thickness, t_{ox}, determines both the current handling and the reliability of the transistors, it must be controlled to within a few percent. For example, the oxide thicknesses of two transistors separated by 20 cm on a wafer must differ by less than a few angstroms, requiring extremely high uniformity across the wafer and hence a slow growth of the oxide. Also, the "cleanness" of the silicon surface under the oxide affects the mobility of the charge carriers and thus the current drive, transconductance, and noise of the transistors.

18.5 ■ Ion Implantation

In many fabrication steps, dopants must be selectively introduced into the wafer. For example, after the lithography sequence of Fig. 18.3 is completed, the n-well is formed by entering dopants into the exposed silicon area. Similarly, the source and drain regions of transistors require selective addition of dopants to the wafer.

The most common method of introducing dopants is "ion implantation," whereby the doping atoms are accelerated as a high-energy focused beam, hitting the surface of the wafer and penetrating the exposed areas [Fig. 18.5(a)]. The doping level (dosage) is determined by the intensity and duration of the implantation, and the depth of the doped region is set by the energy of the beam. As shown in Fig. 18.5, with a high energy, the peak of the doping concentration in fact occurs well below the surface, thereby creating a "retrograde" profile. Such a profile is desirable for the n-well because it establishes a low resistivity near the bottom, reducing susceptibility to latch-up (Sec. 18.8), and a low doping level at the surface, decreasing the S/D junction capacitance of PMOS devices.

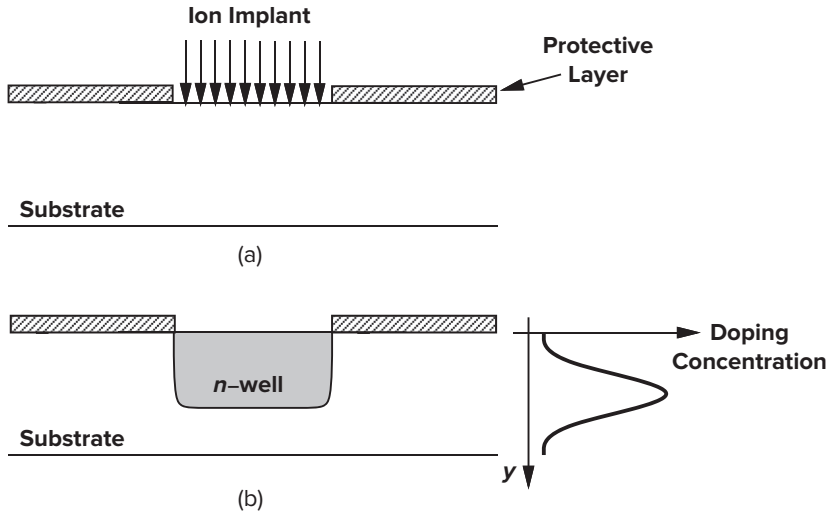


Figure 18.5 (a) Ion implantation; (b) retrograde profile.

Another important application of implantation is to create "channel-stop" regions between transistors. Consider the field oxide and the S/D junctions of M_1 and M_2 in Fig. 18.6(a), assuming that an interconnect line passes on top of the field oxide. Interestingly, the two n^+ regions and the FOX form a MOS transistor having a thick gate oxide and hence a large threshold voltage. Nonetheless, with a sufficiently positive potential on the interconnect line, this transistor may turn on slightly, creating a leakage path between

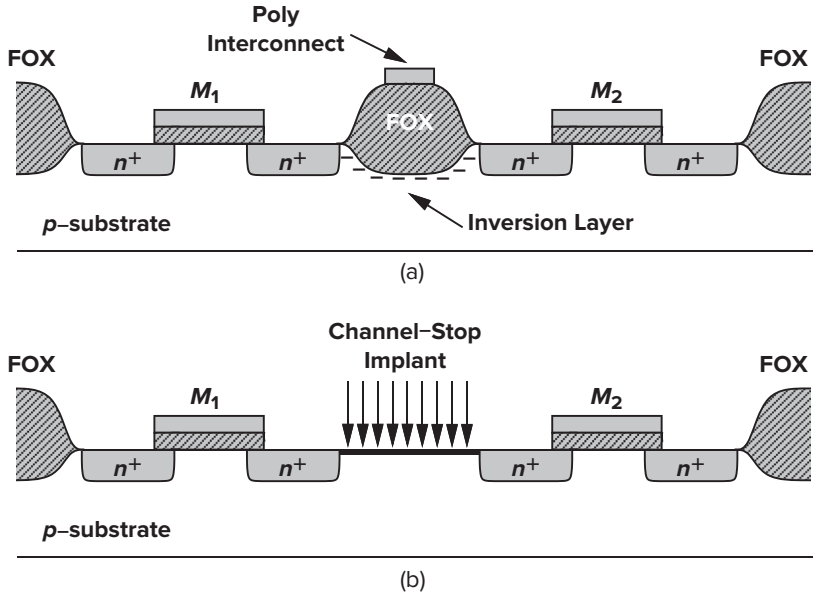


Figure 18.6 (a) Unwanted conduction due to inversion of field area; (b) channel-stop implant.

M_1 and M_2. To resolve this issue, a channel-stop implant (also called a field implant) is performed before the field oxide deposition [Fig. 18.6(b)], thereby raising the threshold voltage of the field oxide transistor to a very large value.

Ion implantation damages the silicon lattice extensively. For this reason, the wafer is subsequently heated to approximately 1, 000°C for 15 to 30 minutes, allowing the lattice bonds to form again. Called "annealing," this operation also leads to diffusion of dopants, broadening the profile in all directions. For example, annealing results in side diffusion of S/D regions, creating overlap with the gate area. The wafer is therefore usually annealed only once, after all implantations have been completed.

An interesting phenomenon in ion implantation is "channeling." As shown in Fig. 18.7(a), if the implant beam is aligned with the crystal axis, the ions penetrate the wafer to a great depth. For this reason, the implant (or the wafer) is tilted by 7–9° [Fig. 18.7(b)], avoiding such an alignment and ensuring a predictable profile. As explained in Chapter 19, this tilt affects the matching of transistors, necessitating precautions in the layout.

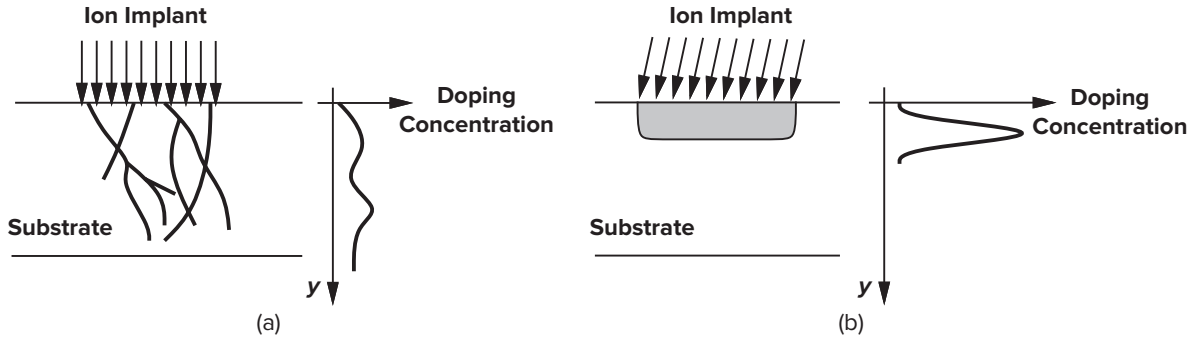


Figure 18.7 (a) Effect of channeling; (b) tilt in implant to avoid channeling.

18.6 ■ Deposition and Etching

As suggested by the structures of Fig. 18.1, device fabrication requires the deposition of various materials. Examples include polysilicon, dielectric materials separating interconnect layers, and metal layers serving as interconnects.

A common method of forming polysilicon on thick dielectric layers is "chemical vapor deposition" (CVD), whereby wafers are placed in a furnace filled with a gas that creates the desired material through a chemical reaction. In modern processes, CVD is performed at a low pressure to achieve more uniformity.

The etching of the materials is also a crucial step. For example, contact windows with very small dimensions, e.g., 0.3 μm $\times$ 0.3 μm, and relatively large depths, e.g., 2 μm, must be etched with high precision. Depending on the speed, accuracy, and selectivity required in the etching step, and the type of material to be etched, one of these methods may be used: (1) "wet" etching, i.e., placing the wafer in a chemical liquid (low precision); (2) "plasma" etching, i.e., bombarding the wafer with a plasma gas (high precision); (3) reactive ion etching (RIE), where ions produced in a gas bombard the wafer.

18.7 ■ Device Fabrication

With the processing operations described in the previous section, we now study the fabrication sequence and device structures in typical CMOS technologies. We consider three categories: active devices, passive devices, and interconnects.

18.7.1 Active Devices

Basic Transistor Fabrication The fabrication begins with a *p*-type silicon wafer approximately 1 mm thick. Following the cleaning and polishing steps, a thin layer of silicon dioxide is grown as a protective coating on top of the wafer [Fig. 18.8(a)]. Next, to create the *n*-wells, a lithography sequence consisting of photoresist deposition, exposure to UV light using the *n*-well mask, and selective etching is carried out, and the *n*-wells are implanted [Fig. 18.8(b)]. The remaining photoresist and oxide layers are then removed [Fig. 18.8(c)].

Recall from the previous section that a field implant and a field oxide growth are necessary in the areas between the transistors. At this point in the sequence, a stack consisting of a silicon oxide layer, a silicon nitride (Si_3N_4) layer, and a *positive* photoresist layer is created. Next, the "active" mask is used for lithography so that only the regions between the transistors are exposed [Fig. 18.8(d)].[2] Subsequently, the channel-stop implant is performed, the photoresist is removed, and a thick oxide layer is grown in the exposed silicon areas, producing the field oxide. The protective nitride and oxide layers are then removed [Fig. 18.8(e)], thereby exposing all areas where transistors are to be formed. In the subsequent diagrams, the channel-stop implant will be omitted for the sake of clarity.

The next step involves the growth of the gate oxide, a critical operation requiring slow, low-pressure CVD [Fig. 18.8(f)]. As explained in Chapter 2, the "native" threshold voltage of the transistors is typically far from the desired value, necessitating a threshold-adjust implant. (The native threshold of both PMOS and NMOS is usually more negative than desired, e.g., $V_{THN} \approx 0$, and $V_{THP} \approx -1$ V.) Such an implant is performed following the growth of the gate oxide, creating a thin sheet of dopants near the surface and making the threshold of both NMOS and PMOS devices more positive than their nominal values.

[2] The *n*-wells are not shown for clarity.

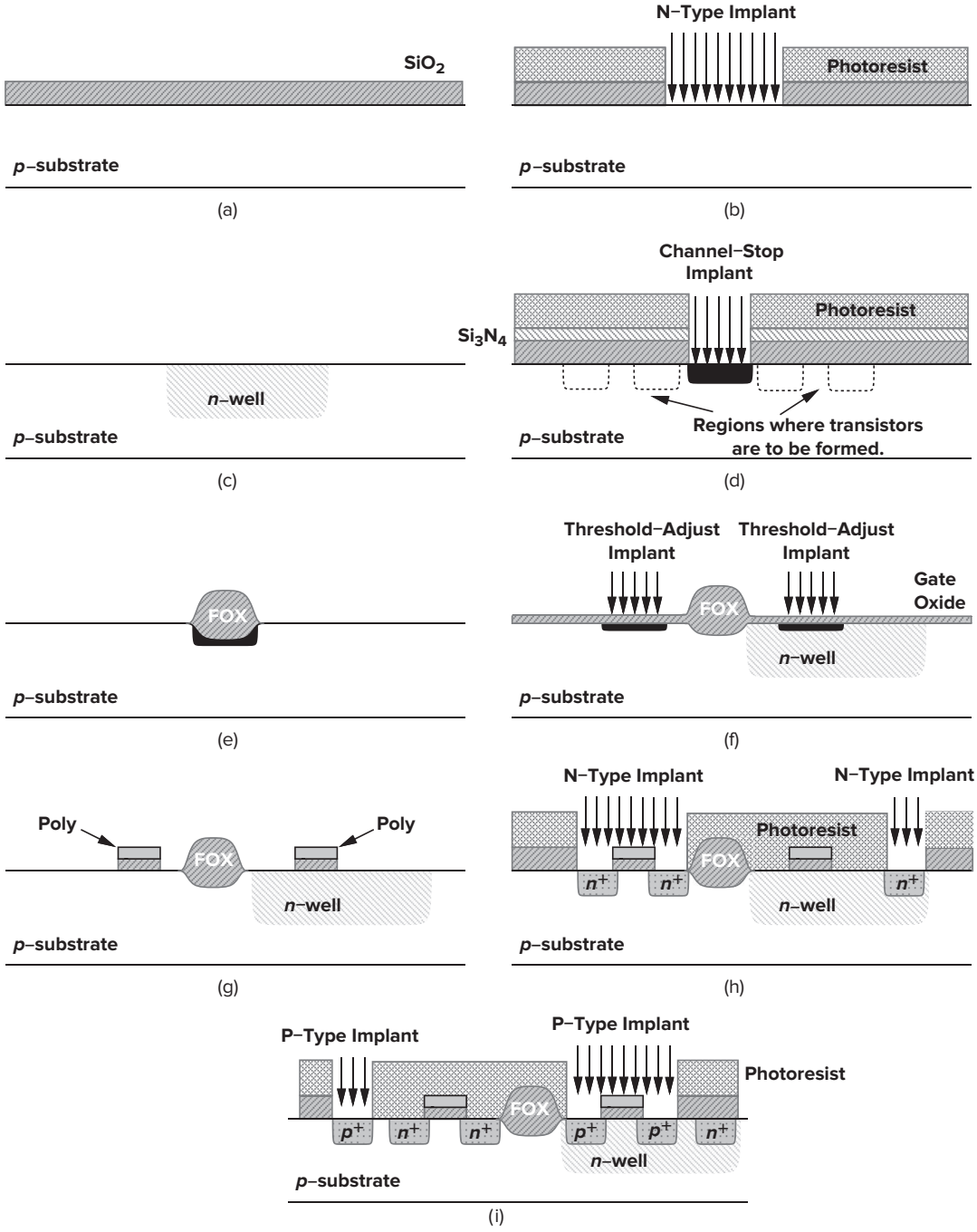


Figure 18.8 Fabrication sequence of MOS devices.

With the gate oxide in place, the polysilicon layer is deposited and the "poly mask" lithography is carried out, resulting in the structure shown in Fig. 18.8(g). We should note that polysilicon is simply noncrystalline ("amorphous") silicon, a property that arises because this layer grows on top of silicon dioxide and hence cannot form a crystal. Since polysilicon serves as a conductor, its amorphous nature is unimportant. To reduce the resistivity of this layer, an additional implant is typically used, yielding a sheet resistance of a few tens of ohms per square.

In the next step, the source/drain junctions of the transistors and the substrate and n-well ties are formed by ion implantation. This step requires a "source/drain mask" and two lithography sequences. As illustrated in Fig. 18.8(h), the first sequence incorporates a negative photoresist, exposing the areas to receive an n^+ implant (the S/D junctions of NMOS transistors and the n-well ties). In the second sequence [Fig. 18.8(i)], the same mask and a positive photoresist are used, exposing the areas to receive a p^+ implant (the S/D junctions of PMOS transistors and the substrate ties). Note that these implants also dope the polysilicon layer, reducing its sheet resistance. This step completes the fabrication of the basic transistors.

The reader may wonder why the source/drain junctions are formed *after* the gate oxide and polysilicon. Suppose, as depicted in Fig. 18.9(a), these junctions were created first. Then, the alignment of the gate poly mask with respect to the S/D areas would become extremely critical. Even if the misalignment were a small fraction of the minimum channel length, a gap might appear between the source (or drain) and the gate area, prohibiting the formation of a continuous channel in the transistor. By contrast, the sequence shown in Fig. 18.8 yields a "self-aligned" structure because the source/drain regions are implanted at precisely the edges of the gate area and a misalignment in lithography simply makes one junction slightly narrower than the other [Fig. 18.9(b)]. Interestingly, the first few generations of CMOS technology were based on the approach shown in Fig. 18.9(a), but it was soon discovered that the self-aligned structure would lend itself to scaling much more easily.

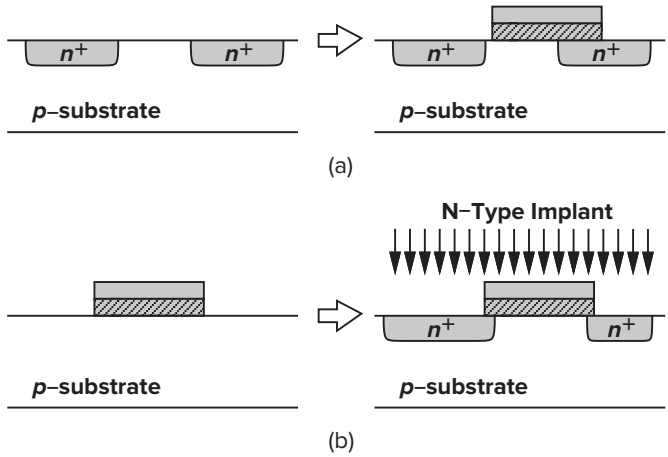


Figure 18.9 (a) Formation of n^+ regions before deposition of poly; (b) self-aligned structure.

Back-End Processing With the basic transistors fabricated, the wafers must next undergo "back-end" processing, a sequence primarily providing various electrical connections on the chip through contacts and wires. The first step in this sequence is "silicidation." Since the sheet resistance of doped polysilicon and S/D regions is typically several tens of ohms per square, it is desirable to reduce their resistance by about an order of magnitude. Silicidation accomplishes this by covering the polysilicon layer and active areas (S/D regions and substrate and n-well ties) with a thin layer of a highly conductive material, e.g., titanium silicide or tungsten. Illustrated in Fig. 18.10, this step in fact begins with creating an "oxide spacer" at

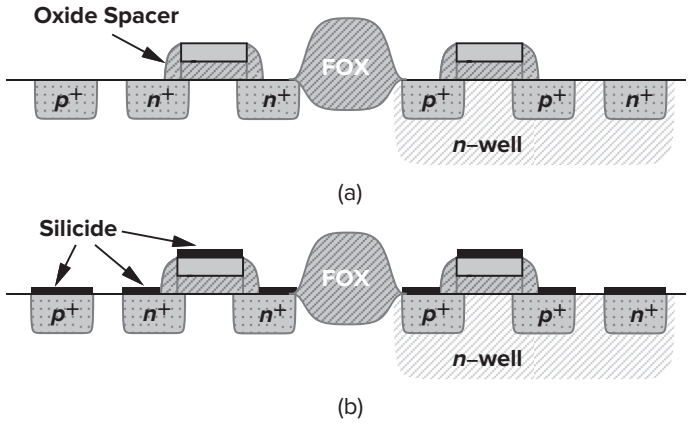


Figure 18.10 (a) Oxide spacers and (b) silicide.

the edges of the polysilicon gate such that the deposition of the silicide becomes a self-aligned process as well.[3] Without the spacer, the silicide layer on the gate may be shorted to that on the source/drain.

The next step in back-end processing is to produce contact windows on top of the polysilicon and active regions. This is carried out by first covering the wafer with a relatively thick (0.3- to 0.5-μm) layer of oxide and subsequently performing a lithography sequence using the "contact mask." The contact holes are then created by plasma etching [Fig. 18.11(a)]. Owing to reliability issues, contacts to the gate polysilicon are not placed on top of the gate area.

Following the contact windows, the first layer of metal interconnect (called "metal 1") (using aluminum or copper) is deposited over the entire wafer. A lithography sequence using the "metal 1 mask" is then carried out, and the metal layer is selectively etched [Fig. 18.11(b)].

The higher levels of interconnect are fabricated using the same procedure [Fig. 18.11(c)]. For each additional metal layer, two masks are required: one for the contact windows and another for the metal itself. Thus, a CMOS process having five layers of metal contains 10 masks for the back end. The contact windows between metal layers are sometimes called "vias" to distinguish them from the first level of contacts to active areas and polysilicon.

We should mention that if a large area must be contacted, many small windows—rather than a large window—are usually used. Dictated by reliability issues, the dimensions of each contact or via are fixed and cannot be decreased or increased by the layout designer. An interesting phenomenon related to large active areas is "contact spiking." If a large contact window allows aluminum to touch the active area, then, as depicted in Fig. 18.12(a), the metal may "eat" and penetrate the doped region, eventually crossing the junction to the bulk and shorting the diode. With small windows, on the other hand, this effect is avoided [Fig. 18.12(b)].

The final step in back-end processing is to cover the wafer with a "glass" or "passivation" layer, protecting the surface against damages caused by subsequent mechanical handling and dicing. After a lithography sequence using the "passivation mask," the glass is opened only on top of the bond pads to allow connection to the external environment (e.g., the package).

18.7.2 Passive Devices

Passive components such as resistors and capacitors find wide usage in analog design, making it desirable to add these devices to standard CMOS technologies. In practice, however, CMOS processes target

[3] Self-aligned silicide is sometimes called "salicide."

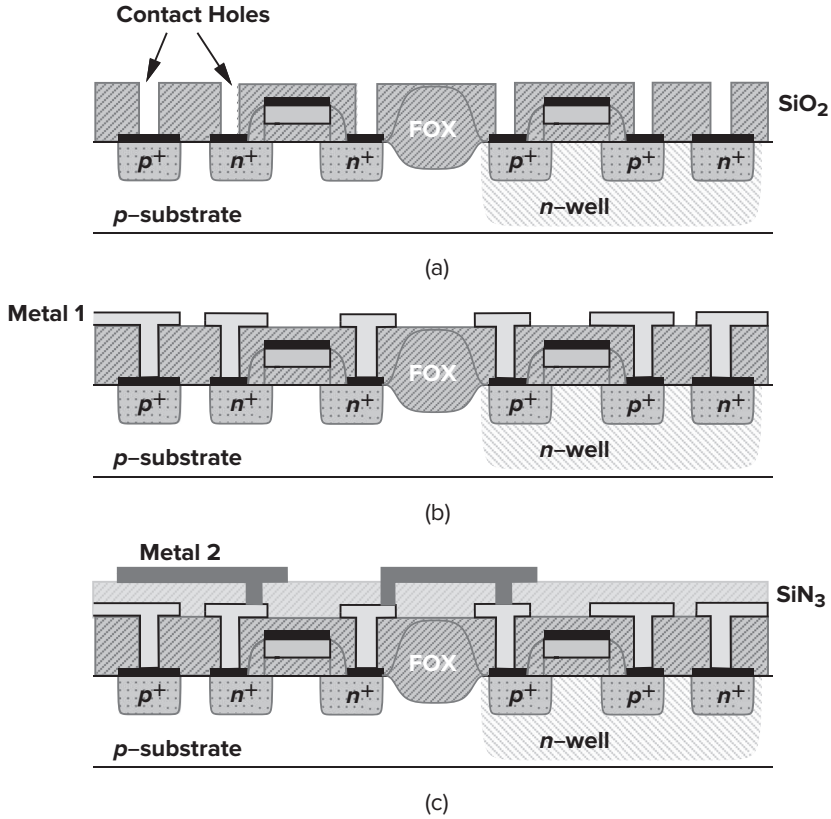


Figure 18.11 Contact and metal fabrication.

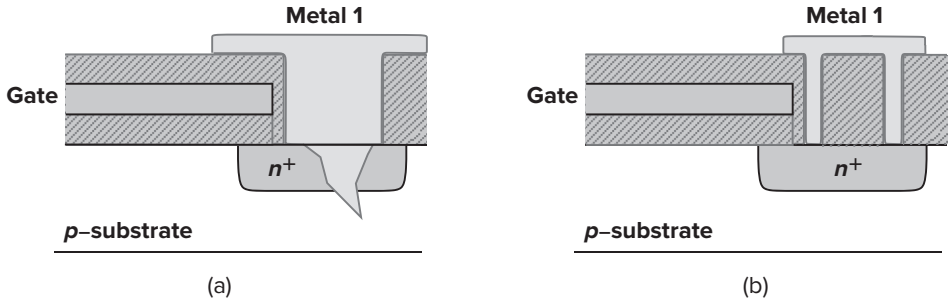


Figure 18.12 (a) Spiking due to large contact areas; (b) use of small contacts to avoid spiking.

primarily digital applications and hence provide only NMOS and PMOS transistors. A new generation of CMOS technology may take one to two years and many iterations before it becomes an "analog process," i.e., one offering high-quality passive devices. If a digital CMOS process is to be used for analog design, we must seek structures that can serve as passive components. The principal issue in using such structures is the *variability* of the component value from wafer to wafer because the process flow does not assume such structures are used in circuits.

Resistors A CMOS process may be modified so as to provide resistors suited to analog design. A common method is to selectively "block" the silicide layer that is deposited on top of the polysilicon, thereby creating a region having the resistivity of the doped polysilicon (Fig. 18.13). This means that the fabrication requires an additional mask and a corresponding lithography sequence. Since the poly doping level is determined by various implants in the process, the resistivity obtained here is not necessarily a target value, but it usually falls in the range of fifty to a few hundred ohms per square. For the same reason, the resistance value may vary by as much as $\pm 20\%$ from wafer to wafer or lot to lot.

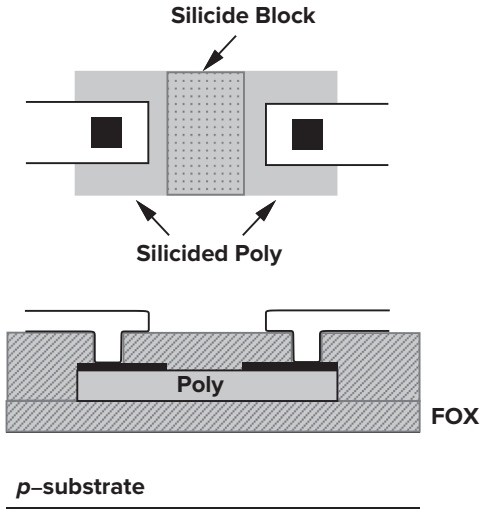


Figure 18.13 Poly resistor using silicide block.

The use of silicide on the two ends of the resistor in Fig. 18.13 results in a much lower contact resistance than that obtained by directly connecting the metal layer to doped polysilicon. This improves both the definition of the resistor value and the matching with identical structures. Also, for a given resistance, poly resistors typically exhibit much less capacitance to the substrate than other types—on the order of 90 aF/μm^2 for the bottom plate capacitance and 100 aF/μm for the fringing capacitance. These resistors are quite linear, especially if they are long. The primary difficulties with silicide-block poly resistors are variability, mask cost, and process complexity.

In a purely digital process, silicided poly, silicided p^+ or n^+ active areas, n-well, and metal layers can be used as resistors. An n-well resistor can be formed as shown in Fig. 18.14, but the n-well resistivity may vary by several tens of percent with process. With typical sheet resistivities of about 1 k$\Omega/\square$, n-well resistors can prove useful where their absolute value is not critical. For example, Fig. 18.15 shows a common-source stage that is biased by means of M_0 and I_0 while employing C_1 to block the dc level of the preceding stage. In order to isolate the signal path from the low impedance (and the noise) introduced by M_0, resistor R_1 is inserted between X and Y. Here, the value of R_1 is not critical so long as it is sufficiently large.

We should mention that, due to the depletion region formed between the n-well and the p-substrate, n-well resistors suffer from both a large parasitic capacitance and significant voltage dependence. Figure 18.16 illustrates a typical case, where one terminal of the n-well resistor is tied to V_{DD}. Since the capacitance to the substrate is distributed (nonuniformly) along the resistor, a lumped model may not be accurate enough, but as a rough approximation, we place half of the total capacitance on each side of the resistor. We also note that as V_{out} varies, so do the width of the depletion region and hence the value of the resistor.

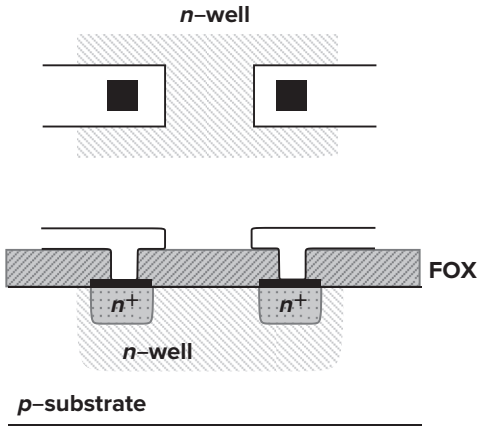


Figure 18.14 Resistor made of n-well.

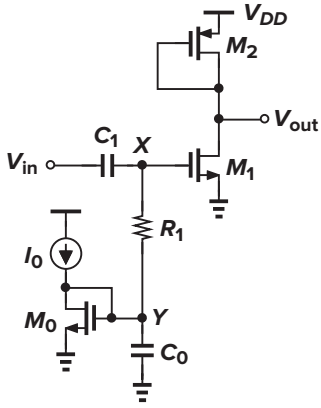


Figure 18.15 Use of an n-well resistor in a coupling network.

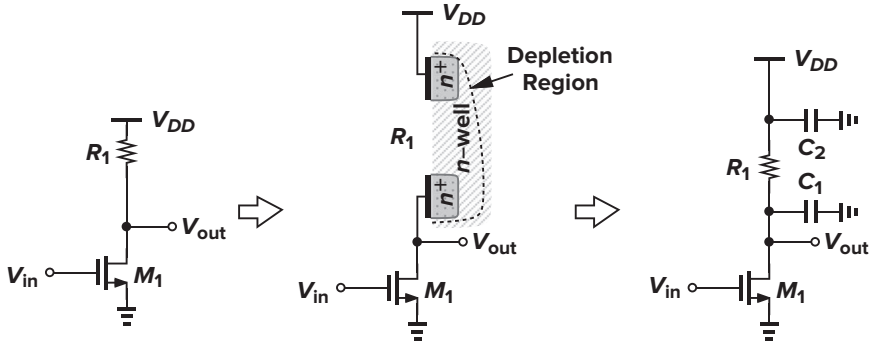


Figure 18.16 Common-source stage using n-well resistors.

The metal layers available in CMOS technologies exhibit sheet resistances on the order of 100 mΩ/□ (for bottom layers) to 30 mΩ/□ (for top layers). Thus, for resistor values common in analog design, metal layers are rarely used.

Capacitors Capacitors prove indispensible in most of today's analog CMOS circuits. Several parameters of capacitors are critical in analog design: parasitic capacitance to the substrate, capacitance per unit area (density), and nonlinearity.

Perhaps the simplest capacitor structure in CMOS technology is that implemented by a MOSFET. Illustrated in Fig. 18.17(a), the device has a capacitance that varies from a small value at low voltages (where no channel exists and the equivalent capacitance is the series combination of the oxide capacitance and the depletion region capacitance) to a large value (C_{ox}) if the voltage difference exceeds V_{TH}. Since the gate oxide is typically the thinnest layer in the process, MOS capacitors biased in strong inversion are quite dense, saving substantial area if large values are required. For the same reason, the bottom-plate parasitic, i.e., that due to drain and source junctions, is a relatively small percentage of the gate capacitance—typically 10 to 20%.

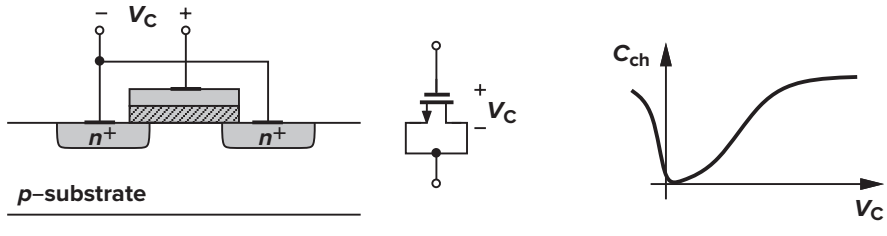


Figure 18.17 (a) MOSFET configured as a capacitor; (b) nonlinear C/V characteristic.

Unfortunately, the voltage dependence of MOS capacitors, even in strong inversion, makes the structure less attractive for precision charge transfer.

▶ Example 18.1

Consider the multiply-by-two amplifier of Sec. 13.3.3, shown in Fig. 18.18(a) as an implementation using a MOS capacitor C_1 and a linear capacitor C_2. Explain how the output voltage in the amplification mode is distorted.

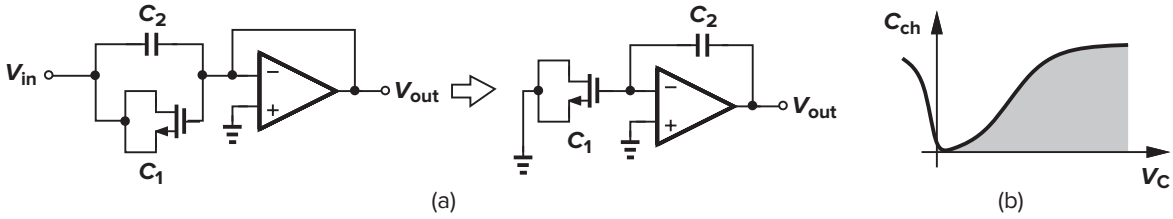


Figure 18.18 Precision multiply-by-two circuit using a MOS capacitor.

Solution

Suppose for simplicity that V_{in} is below ground by more than V_{TH}, so that the NMOS capacitors are in strong inversion during sampling. As the circuit enters the amplification mode, the voltage across C_1 approaches zero and the total charge stored on C_1 is transferred to C_2. How much is this charge? If C_1 were linear, we would have $Q = C_1 V$, but here we must write $dQ = C_1 dV$. Thus, as shown in Fig. 18.18(b), the total transferred charge when the voltage across the capacitor goes from V_{in} to zero is equal to the area under the C/V characteristic, a value substantially less than that in the linear case. The output voltage is then given by

$$V_{out} \approx V_{in} + \frac{1}{C_2}\int_0^{V_{in}} C_1 dV \tag{18.1}$$

◀

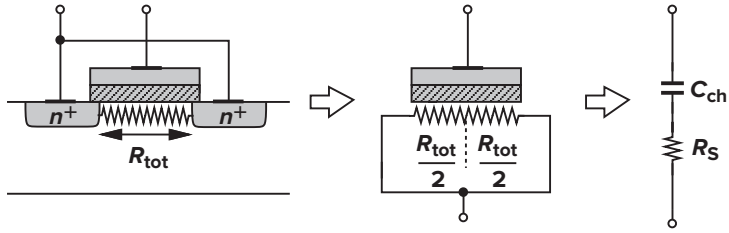


Figure 18.19 Channel resistance of MOS capacitor.

Another issue related to MOS capacitors is their series resistance, an effect arising from the gate material and, more important, the channel resistance. Assuming that proper layout minimizes the gate resistance, we view the channel resistance as shown in Fig. 18.19, estimating the equivalent series resistance as $(R_{tot}/2)\|(R_{tot}/2) = R_{tot}/4$, where $R_{tot} = [\mu C_{ox}(W/L)(V_{GS} - V_{TH}]^{-1}$. The intrinsic time constant of the capacitor is therefore equal to

$$\tau = \frac{R_{tot}}{4} C_{ch} \tag{18.2}$$

$$= \frac{1}{4\mu C_{ox}(W/L)(V_{GS} - V_{TH})} \cdot WLC_{ox} \tag{18.3}$$

$$= \frac{L^2}{4\mu(V_{GS} - V_{TH})} \tag{18.4}$$

In reality, the distributed nature of the resistance and the capacitance along the channel results in a time constant equal to one-third of that given above [2]. Another figure of merit for such a capacitor is $Q = [1/(C\omega)]/R_S$. As a rule of thumb, we choose $R_S < 0.1/(C\omega)$.

Equation (18.4) indicates that for a given overdrive, to minimize the series resistance of a MOS capacitor, L must be minimized. Consequently, MOS capacitors are usually designed as a parallel combination of wide, short devices rather than as a *square* block (Fig. 18.20). The penalty is a higher junction capacitance to the substrate and somewhat greater area.

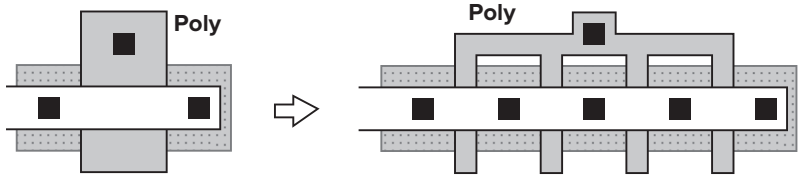


Figure 18.20 Use of wide, short MOS fingers to reduce channel resistance.

In applications requiring linear capacitors, a "sandwich" of conductive layers can be formed in CMOS technology. Shown in Fig. 18.21 is an example, where the capacitance between metal layers is exploited to increase the density. Since the dielectrics between the layers are relatively thick, this structure still requires a large area. More important, the bottom-plate parasitic (e.g., the capacitance between the lowest layer and the substrate in Fig. 18.21) is significant, about 5 to 10% of the total interplate capacitance. This structure is studied in detail in Chapter 19.

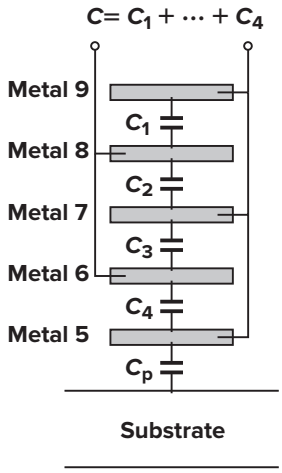


Figure 18.21 Linear capacitor made of native conductive layers.

▶ **Example 18.2**

An amplifier with an input capacitance of C_{in} is to be ac-coupled to a preceding stage having an output resistance R_{out}. Considering both of the topologies depicted in Fig. 18.22 and allowing a maximum signal attenuation of 20%, determine the minimum value of the coupling capacitor and the resulting time constant if $C_P = 0.5C_C$ or $C_P = 0.2C_C$.

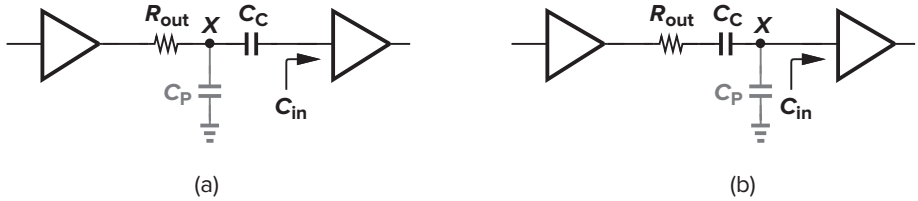


Figure 18.22

Solution

In Fig. 18.22(a), the attenuation is given by $A_v = C_C/(C_C + C_{in})$, yielding $C_C \geq 4C_{in}$ for a 20% signal loss. The total capacitance seen from node X to ground is therefore equal to $C_P + C_C C_{in}/(C_C + C_{in}) = C_P + 0.8C_{in}$. It follows that the time constant is $2.8R_{out}C_{in}$ for $C_P = 0.5C_C$ and $1.6R_{out}C_{in}$ for $C_P = 0.2C_C$.

In Fig. 18.22(b), C_P itself attenuates the signal: $A_v = C_C/(C_C + C_{in} + C_P)$, indicating that no value of C_C can yield a signal loss of 20% if $C_P \geq 0.25C_C$.

These calculations yield two important results. First, the topology of Fig. 18.22(a) is generally preferable. Second, the addition of a coupling capacitor, e.g., to isolate the bias levels, substantially degrades the speed. ◀

18.7.3 Interconnects

The performance of today's complex integrated circuits heavily depends on the quality of the available interconnects, requiring more metal layers in new generations of the technology.[4] Proper modeling of interconnects in a high-performance circuit is still a topic of active research, but our objective is to provide a basic understanding of the interconnect issues.

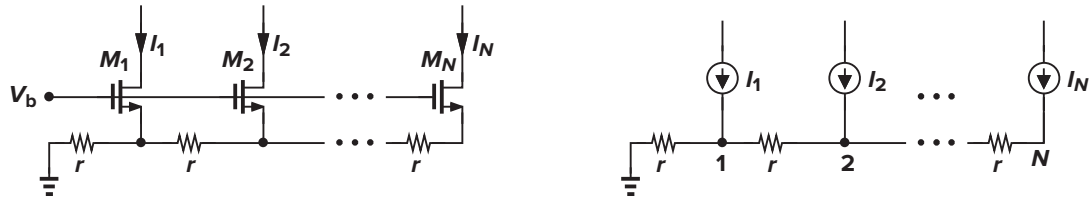
[4] At the time of this writing, five layers of metal are in production.

Two properties of interconnects, namely, series resistance and parallel capacitance, affect the performance, often calling for iteration between layout and circuit design. The series resistance becomes especially problematic in supply and ground lines, creating dc and transient voltage drops. Also, for long signal lines, the distributed resistance and capacitance of the wire may result in a significant delay.

The resistance of metal wires can be easily estimated at low frequencies, at which skin effect is negligible. Typical sheet resistances are 30 mΩ/□ for the topmost (thickest) layer and 100 mΩ/□ for lower layers. The finite resistance of wires influences the choice of line widths for high-current interconnects such as supply and ground buses, as illustrated by the following example.

▶ Example 18.3

A D/A converter incorporates N equal current sources implemented as NMOS devices, each having an aspect ratio of W/L [Fig. 18.23(a)]. Assuming that the interconnect between every two consecutive current sources has a small resistance, r, estimate the mismatch between I_N and I_1.



(a) (b)

Figure 18.23 Effect of ground resistance in a D/A converter.

Solution

If r is sufficiently small, the circuit can be modeled as shown in Fig. 18.23(b), where $I_1 \approx I_2 \approx \cdots \approx I_N = I$. The voltage at node N is obtained by superposition of currents:

$$V_N = Ir + I(2r) + \cdots + I(Nr) \tag{18.5}$$

$$= \frac{N(N+1)}{2} Ir \tag{18.6}$$

If V_N is relatively small, the assumption that $I_1 \approx I_2 \approx \cdots \approx I_N$ used in the above calculation is reasonable and M_1–M_N exhibit roughly equal transconductances. Thus,

$$I_N = I - g_m V_N \tag{18.7}$$

$$= I - g_m r \frac{N(N+1)}{2} I \tag{18.8}$$

$$= I \left[1 - g_m r \frac{N(N+1)}{2} \right] \tag{18.9}$$

Since $V_1 \approx N \cdot I \cdot r$, we have $I_1 = I - g_m N \cdot I \cdot r$, and the relative mismatch between I_1 and I_N is

$$\left| \frac{I_1 - I_N}{I} \right| = g_m r \frac{N(N-1)}{2} \tag{18.10}$$

The key point here is that the error grows in proportion to N^2. The ground bus must therefore be sufficiently wide to minimize r.

◀

Another factor determining the width of interconnects is "electromigration." At high current densities, the aluminum atoms in a wire tend to "migrate," leaving a void that eventually (after some years of operation) grows to a discontinuity. For this reason, long-term reliability considerations restrict the maximum current density of interconnects. As a rule of thumb, a current density of 2 mA per micron of width is acceptable, but the actual value varies according to the thickness of the metal. Also, for transient currents, the peak value may be quite a lot higher.

The problem of interconnect capacitance is much more complicated. We begin with a single wire on top of a substrate (Fig. 18.24), identifying a "parallel-plate" capacitance and a "fringe" capacitance. For narrow lines, the two are comparable.

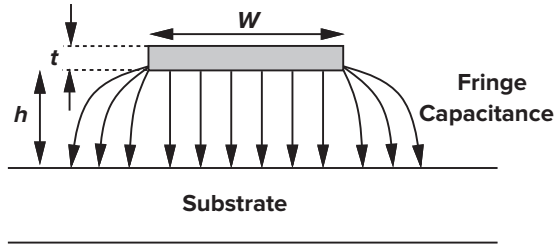


Figure 18.24 Parallel-plate and fringe capacitance of an interconnect.

A simple empirical relationship for calculating the total wire capacitance per unit length on top of a conducting substrate is

$$C = \epsilon \left[\frac{W}{h} + 0.77 + 1.06 \left(\frac{W}{h} \right)^{0.25} + 1.06 \left(\frac{t}{h} \right)^{0.5} \right] \tag{18.11}$$

where W, h, and t denote the dimensions shown in Fig. 18.24 [3]. For typical dimensions, this equation predicts the capacitance with a few percent of error.

While upper levels of metal in a process exhibit less capacitance per unit width and length, their minimum allowable width is usually greater than that of the lower layers. Thus, the minimum capacitance for a given length may be only slightly smaller for the topmost layer(s). Table 18.1 depicts typical values of minimum widths and parallel-plate and fringe capacitances (to the substrate) in a four-metal 0.25-μm process.

Table 18.1 Minimum widths and capacitances of interconnects in a 0.25-μm technology.

	Poly	Metal 1	Metal 2	Metal 3	Metal 4
Minimum Width (μm)	0.25	0.35	0.45	0.50	0.60
Bottom–Plate Capacitance (aF/μm^2)	90	30	15	9.0	7.0
Fringe Capacitance (Two Sides) (aF/μm)	110	80	50	40	30

Wires also suffer from parallel and fringe capacitances between them. Illustrated in Fig. 18.25, this effect is difficult to quantify for a complex layout, often necessitating the use of computer programs. In

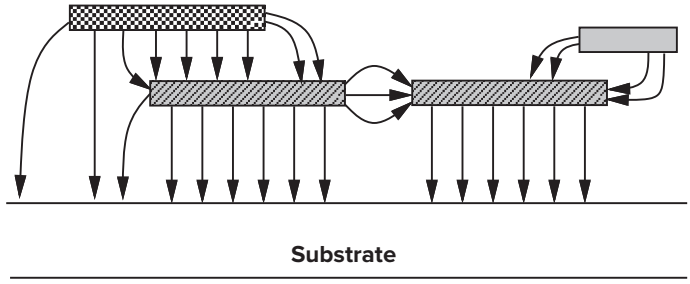


Figure 18.25 Complex interconnect structure.

practice, the capacitances between the layers are calculated by "electromagnetic field solvers," measured experimentally, and tabulated in the process design manual.

18.8 ■ Latch-Up

Owing to manufacturing difficulties, the first few generations of MOS technologies provided only NMOS devices. In fact, many of the early microprocessors and analog circuits were fabricated in NMOS processes, but they consumed substantial power. The advent of CMOS technology was motivated by the zero static power dissipation of CMOS logic—although CMOS devices required a greater number of masks and fabrication steps. Another issue that did not exist in NMOS implementations but arose in CMOS circuits was latch-up.

Consider the NMOS and PMOS devices shown in Fig. 18.26(a). Recall from Chapter 12 that a parasitic *pnp* bipolar transistor, Q_1, is associated with the PFET, the *n*-well, and the substrate. By the same token, a parasitic *npn* device, Q_2, can be identified in conjunction with the NFET. We make two observations: (1) the base of each bipolar transistor is inevitably tied to the collector of the other; and (2) owing to the finite resistance of the *n*-well and the substrate, the bases of Q_1 and Q_2 see a nonzero resistance to V_{DD} and ground, respectively. The parasitic circuit can therefore be drawn as in Fig. 18.26(b), revealing a *positive* feedback loop around Q_1 and Q_2. In fact, if a current is injected into node X such that V_X rises, then I_{C2} increases, V_Y falls, $|I_{C1}|$ increases, and V_X rises further. If the loop gain is greater than or equal to unity, this phenomenon continues until both transistors turn on completely, drawing an enormous current from V_{DD}. We say that the circuit is latched up.

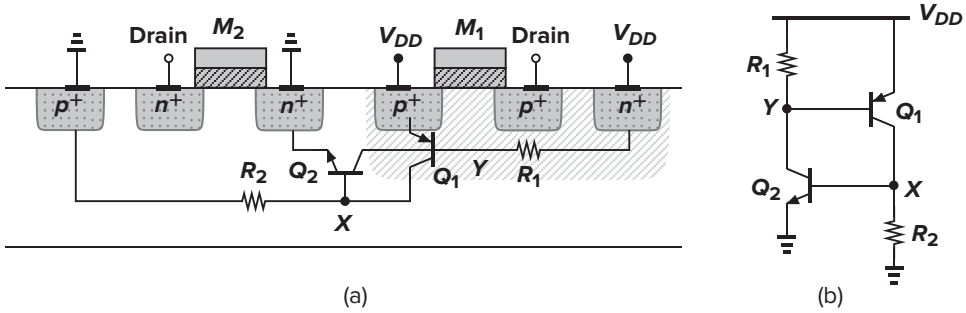


Figure 18.26 (a) Parasitic bipolar transistors in a CMOS process; (b) equivalent circuit.

The initial current required to trigger latch-up may be produced by various sources in an integrated circuit. For example, in Fig. 18.26(a), the bases of Q_1 and Q_2 are capacitively coupled to the drains of M_1 and M_2, respectively. A large voltage swing at the drains can therefore inject a significant displacement current into the *n*-well or the substrate, initiating latch-up.

A common case of latch-up occurs with the use of large digital output buffers (inverters). These circuits inject high currents into the substrate through the large drain junction capacitance of the transistors and by forward-biasing the source-bulk junction diodes. The latter arises because of the substantial transient voltages produced across the bond wires connected to the ground (Chapter 19).

In order to prevent latch-up, both process engineers and circuit designers take precautions to ensure that the loop gain of the equivalent circuit shown in Fig. 18.26(b) remains well below unity. Proper choice of the doping levels and profiles as well as layout design rules ensure a low value for both the parasitic resistances and the current gain of the bipolar transistors. Furthermore, the layout of the circuit incorporates substrate and n-well contacts with sufficiently small spacing to minimize the resistance. The design manual of each technology typically provides an extensive set of layout rules recommended for latch-up prevention.

References

[1] C. Kaya et al., "Polycide/Metal Capacitors for High Precision A/D Converters," *IEDM Dig. of Tech. Papers*, pp. 782–785, December 1988.

[2] P. Larsson, "Parasitic Resistance in an MOS Transistor Used as On-Chip Decoupling Capacitor," *IEEE J. Solid-State Circuits*, vol. 32, pp. 574–576, April 1997.

[3] E. Barke, "Line-to-Ground Capacitance Calculations for VLSI: A Comparison," *IEEE Trans. on Commputer-Aided Design*, vol. 7, pp. 195–298, February 1988.

Problems

Unless otherwise stated, in the following problems, use the device data shown in Table 2.1 and assume that $V_{DD} = 3\,\text{V}$ where necessary. Also, assume that all transistors are in saturation.

18.1. A MOS technology is designed to provide only n-type transistors and two metal layers. Sketch the fabrication steps and determine the minimum number of masks required in this technology.

18.2. During a threshold-adjust implant, the wafer was not tilted, leading to severe channeling. Explain whether the resulting threshold voltage is higher or lower than the target value.

18.3. The circuits of Fig. 18.27 have been fabricated with a longer-than-expected gate oxidation cycle. If the threshold voltages are still equal to the desirable value, sketch V_{out} versus V_{in} and compare the results to the target case.

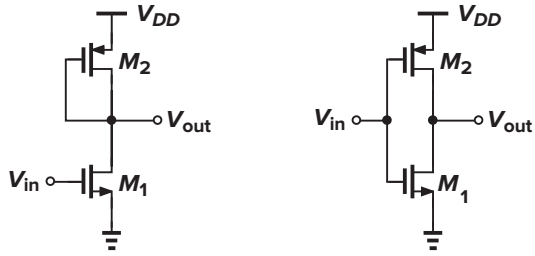


Figure 18.27

18.4. The circuits of Fig. 18.27 have been fabricated without a threshold-adjust implant. Sketch V_{out} versus V_{in} and compare the results to the target case.

18.5. Due to a layout error, the circuit shown in Fig. 18.28 suffers from contact spiking in one of the junctions. Identify the faulty junction if (a) the voltage gain is higher than expected, (b) the output voltage is near V_{DD}.

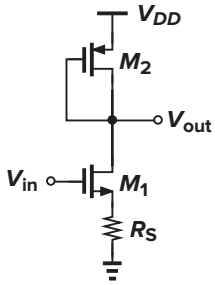


Figure 18.28

18.6. An NMOS cascode current source used in a large circuit exhibits a substantially lower output impedance than expected. Determine which fabrication error may have led to this effect: (a) channeling during S/D implant, (b) omission of the channel-stop implant, or (c) insufficient gate-oxide growth.

18.7. An NMOS cascode current source has a zero output current. If a single (small) lithography misalignment has caused this error, determine in which fabrication step(s) this may have occurred.

18.8. A differential pair using an active current mirror as load suffers from a low small-signal voltage gain. If the bias current is equal to the target value, determine which fabrication error may have led to this effect: (a) heavy n-well implantation, (b) heavy threshold-adjust implantation, or (c) long gate oxidation cycle.

18.9. The switched-capacitor amplifier of Fig. 18.29 exhibits a large gain error. If the bias current of the op amp is equal to the desired value, which fabrication error is likely to have happened: (a) heavy threshold-adjust implantation, (b) very heavy doping in the bottom plate of C_1 (placed at node P), or (c) channeling during the S/D implantation?

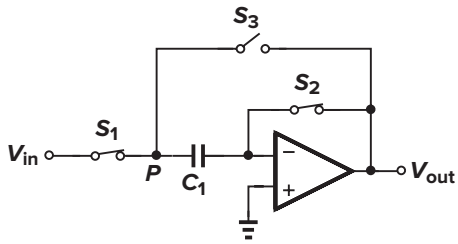


Figure 18.29

18.10. In Fig. 18.30, the digital circuit draws large transient currents from V_{DD}. Without M_1, the inductor L_b would sustain a large transient voltage $L_b dI_{DD}/dt$. Transistor M_1 with $W/L = 100/0.5$ is added to suppress this effect.

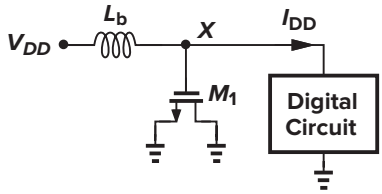


Figure 18.30

(a) Calculate the equivalent series resistance of M_1.
(b) Calculate the maximum value of L_b that results in a critically-damped response at node X. Model the digital circuit by a transient current source.

18.11. In the circuit of Fig. 18.23, $V_b = 1.2$ V, $N = 32$, and $(W/L)_{1-N} = 20/0.5$. Determine the maximum value of r for a maximum current mismatch of 1%.

18.12. Suppose that in Eq. (18.11), $t = 1$ μm and $h = 3$ μm. For what value of W are the parallel-plate and fringe capacitances equal? What if $h = 5$ μm?